www.csvtuonline.com

322712(22)

B. E. (Seventh Semester)
EXAMINATION, Nov.-Dec., 2014
(Old Scheme)
(Branch : CSE)

PARALLEL PROCESSOR AND COMPUTING

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : (i) All questions are compulsory.

(ii) Part (a) is compulsory and carries 2 marks each.

(iii) Part (b), (c) and (d) carry 7 marks each, out of which any *two* have to be attempted.

1. (a) How parallel processing mechanism can be achieved in uniprocesser computer ? Explain it. 2

(b) Explain the FLYNN's taxonomy to classify computer with appropriate diagram. 7

(c) Write Amdahl's law for fixed workload. Explain with complete derivation and draw its graph also. 7

(d) Consider the execution of an object code with 200000 instructions on a 40 MHz processor. The program consists of four major type of

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

instructions. The instruction mix and the no. of cycles needed for each instruction are given below. 7

Instruction Type	CPI	Instruction Mix.
(1) Arithmetic and logic	1	60%
(2) Load	2	18%
(3) Branch	4	12%
(4) M/O reference with cache	8	10%

(i) Calculate the average CPI when the program is executed on a uniprocessor.

(ii) Calculate the corresponding MIPS rate based on the CPI obtained in part (a).

2. (a) Write the properties of parallelism. 2

(b) Explain different types of vector instructions in detail. 7

(c) What are pipelining hazards ? Explain instruction format of cray-1. 7

(d) Performance serial and parallel process for the given condition : 7

P1 : $C = D \times E$

P2 : $M = C_T + C$

P3 : $A = B + C$

P4 : $C = L + M$

P5 : $F = C_T \div E$

www.csvtuonline.com

www.csvtuonline.com

3. (a) Write the difference between static network and dynamic network. 2

(b) Explain wormhole routing along with its methods. 7

(c) Explain perfect shuffle and inverse perfect shuffle network with their application. 7

(d) Describe butterfly, wrapped butterfly and hypercube networks. 7

4. (a) What do you understand by Emulation ? 2

(b) Describe multithreading in multiprocessors. Also explain distributed shared memory. 7

(c) Describe defect level, fault level and error level. 7

(d) Explain parallel file system. 7

5. (a) What do you understand by Resource dependency in parallel system ? 2

(b) Describe MIN-BASED BBN Butterfly in detail. 7

(c) Explain CC-NUMA Stanford dash in detail. 7

(d) Write short notes on the following : 7

(i) MIMD Machines

(ii) CRAY-y-MP

www.csvtuonline.com

www.csvtuonline.com